

S ALAMEEN

Electronics & Communication Engineer

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PROFILE SUMMARY

Final-year ECE graduate (8.69 CGPA) with hands-on internship experience in NPI process engineering at Foxconn, and project experience across embedded systems, VLSI/FPGA design, and industrial automation. Strong fundamentals in electronics, digital design, microcontroller programming, and manufacturing quality processes. Seeking an entry-level role in electronics, embedded systems, or core engineering.

CORE COMPETENCIES

Analog & Digital Electronics • Microcontroller Programming • Communication Protocols (UART, SPI, I2C) • Embedded C / C++ • FPGA & Digital Design (Verilog) • SCADA & PLC Fundamentals • Root Cause Analysis & Failure Analysis • Quality Inspection & Process Improvement

EDUCATION

B.E. Electronics & Communication Engineering

Ramanathapuram, Tamil Nadu

Mohamed Sathak Engineering College, Kilakarai — CGPA: 8.69 / 10

INTERNSHIP EXPERIENCE

NPI Process Engineering Intern — Foxconn Technology India Pvt. Ltd.

Jul 2026

- Supported New Product Introduction (NPI) builds by monitoring production line activities and assembly process verification.
- Performed failure analysis and Root Cause Analysis (RCA) on production defects, contributing to reduced rework and downtime.
- Conducted quality inspections and identified manufacturing issues in real time during production runs.
- Prepared daily production and failure-analysis reports in Microsoft Excel for cross-functional review.
- Collaborated with production and testing teams to improve process efficiency and product quality.

PROJECTS

IoT-Based Smart Distance Monitoring System (ESP8266)

- Developed Embedded C firmware on ESP8266 and interfaced an ultrasonic sensor for real-time distance monitoring and alerts.

FPGA-Based Image Processing & Digital Circuit Design

- Implemented image-processing algorithms in Verilog on Xilinx Vivado for hardware-level processing.
- Designed and simulated analog and digital circuits in Cadence for performance and timing analysis.

Ultrasonic Radar Simulation System

- Built a servo-based ultrasonic scanning system with real-time data visualization over UART communication.

Clap-Activated Light Control System

- Designed a sound-controlled automation circuit using Arduino, a sound sensor, and a relay module with debouncing logic.

TECHNICAL SKILLS

Programming & Firmware: Embedded C, C++, Microcontroller Programming (Arduino, ESP8266), STM32 (familiar — architecture, STM32CubeIDE)

Embedded & Design Tools: Arduino IDE, Xilinx Vivado (FPGA), Cadence, Serial & Hardware Debugging

Industrial & Automation: PLC Fundamentals, SCADA Fundamentals, HMI Basics, Process Automation, Industrial Communication (Basic)

Other Tools: MATLAB, Python, Microsoft Excel, Cisco Packet Tracer (Networking)

Engineering Practices: Root Cause Analysis (RCA), Failure Analysis, Process Improvement, Quality Inspection

CERTIFICATIONS

- Networking — Hands-on with Cisco Packet Tracer (Routing, VLAN, IP Addressing, DHCP, TCP/IP)
- Embedded Systems Design — Microcontroller Fundamentals & Embedded Software Development
- ASIC Design & FPGA — RTL Design (Verilog), Synthesis, Static Timing Analysis, Floorplanning, Routing, Physical Verification (SCL 180nm)
- VLSI & RISC-V — Verilog HDL, VLSI SoC Design, RISC-V RV32I Base ISA
- Digital Design using Xilinx Vivado — Chips to Startup (C2S) Programme
- Industrial Automation — LinkedIn Learning
- SCADA: Collect, Analyze & Visualize Data for Industrial Automation — LinkedIn Learning